AAMIR TALAL

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EDUCATION

CY Tech, Paris *2026 – 2029*

Engineering Cycle (M.Eng), Computer Science – **Top 5% of the cohort**

– **Relevant coursework:** Machine Learning, Deep Learning, Optimisation, Probability & Statistics, Linear Algebra, Algorithms & Data Structures, Databases

CY Tech, Paris *2024 – 2026*

Integrated Preparatory Cycle (MPSI-equivalent): advanced mathematics, physics, computer science

– **Relevant coursework:** Real Analysis, Linear Algebra, Probability, Differential Equations, Algorithms in C & Python, Numerical Analysis

EXPERIENCE

Safran Aircraft Engines *2026 – 2029*

AI Engineer, Applied Research (Apprenticeship) *Montereau-sur-le-Jard*

- Design and deploy LLM agents that automate and digitalise sales administration (ADV) processes
- Build and evaluate agentic workflows that ground generative AI in existing business tools and data
- Work directly with business teams to identify, scope and prioritise automation use cases

APPLIED RESEARCH PROJECTS

Remaining Useful Life Prediction with Calibrated Uncertainty *2026*

Predictive maintenance on NASA C-MAPSS turbofan engines

- Reproducible RUL study on all four C-MAPSS subsets (FD001–FD004: up to 6 operating conditions, 2 fault modes), piecewise-linear target capped at 125 cycles, 30-cycle sensor windows
- Leakage-free protocol: regime-wise sensor normalisation (k-means on operating settings), statistics fitted on training engines only, held-out engines for early stopping, official test set used once
- Benchmarked Ridge and gradient boosting against a 5-member 1D-CNN deep ensemble trained with Gaussian NLL, decomposing aleatoric and epistemic uncertainty; scored with RMSE and the asymmetric NASA score
- Gradient boosting best at 12.4–14.1 test RMSE; the ensemble's 90% intervals cover 88–93% of test engines; CNN reaches 14.6 RMSE on FD002 vs. 22.4 reported for the same backbone (Li et al., 2018)
- Stack: Python, PyTorch, scikit-learn, pandas

Legible: speaker-attributed grounding for legal LLMs *2026*

Winner, Mistral×Stanford Law School Hackathon

- Designed the System 1→System 2 engine: rule-based zoning and a calibrated classifier (Jev) attribute each sentence of a French court ruling to its speaker with a confidence score, conditioning a Mistral LLM that answers with paragraph-level citations
- On the team's 25-ruling benchmark, 4× fewer misattributed answers than a baseline LLM (11 vs 44 out of 224)
- Stack: Python, FastAPI, Mistral API, MCP; live demo: legible-or4n.onrender.com

MiroPolis: multi-agent LLM simulation of parliament *2026*

Top Project, National Assembly Hackathon

- Built a multi-agent digital twin (OASIS) in which LLM deputies, seeded with the chamber's real composition, groups and programmes, debate, amend and vote on a bill
- Compared fine-tuned agents with persona-orchestrated pre-trained LLMs, and pivoted to the latter as faster and more realistic

TECHNICAL SKILLS

Machine Learning	Deep ensembles, uncertainty estimation, 1D-CNNs, gradient boosting, PyTorch
AI / LLM	LLM agents, RAG, prompt engineering, LLM evaluation, MCP, LangChain, Hugging Face
Data	SQL, pandas, NumPy, scikit-learn, data pipelines, n8n
Languages	Python, C, SQL, JavaScript, Shell/Bash, PHP
Spoken	French (native), English (C1), Urdu (C1)

HACKATHONS & ACHIEVEMENTS

- Winner, Mistral×Stanford Law School Hackathon, and winner of the Legora track
- Top Score, IBM×Airbus Hackathon (Télécom Paris)
- Top Project, French National Assembly Hackathon
- Top 20 out of 300, Anthropic×Hugging Face Hackathon
- ML algorithms and maths implemented from scratch on Deep-ML; 50+ LeetCode problems solved

COMMUNITY & INTERESTS

- Content creator, Intellectuel Athlète (personal development / tech): 14,000+ followers, 2–3M monthly views
- Volunteering: 50+ weekly humanitarian outreach rounds (maraudes), Au cœur de la précarité; created an Arabic-learning web app (1,000+ users in 24 h)
- Sport: Brazilian jiu-jitsu and grappling (competition), running